

SIMATS ENGINEERING
ASSESSMENT TOOL III— Interactive Dashboard Project

Course: Data Handling and Visualization	Code: DSA06	CO: CO5
Duration: 180 min	Total Marks: 100	Reg No : 192324298 Name: Akash S

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods, applied through the design of interactive dashboards. (BL4)

Activity Tool : Interactive Dashboard Project

Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Criteria

Criteria	Dashboard Design and Understanding 25%	Visual Analysis and Interpretation 25%	Application of Dashboard Design Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Interactive Dashboard Project — Assessment Tool

Tableau Practical Questions with Datasets

Instructions: Answer all five questions. Use Tableau to construct the required interactive dashboards, analyze the patterns, compare design choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Building a Sales Performance Dashboard

A retail company records regional sales performance across product categories. Using Tableau, connect to the dataset below and build an interactive dashboard containing: a bar chart of Sales by Region, a line chart of Sales trend by Month, and a map view of Sales by Region. Add a Region filter that acts as a global filter across all views. Analyze which region and category combinations drive the highest sales, and explain how the dashboard layout supports quick comparison across regions.

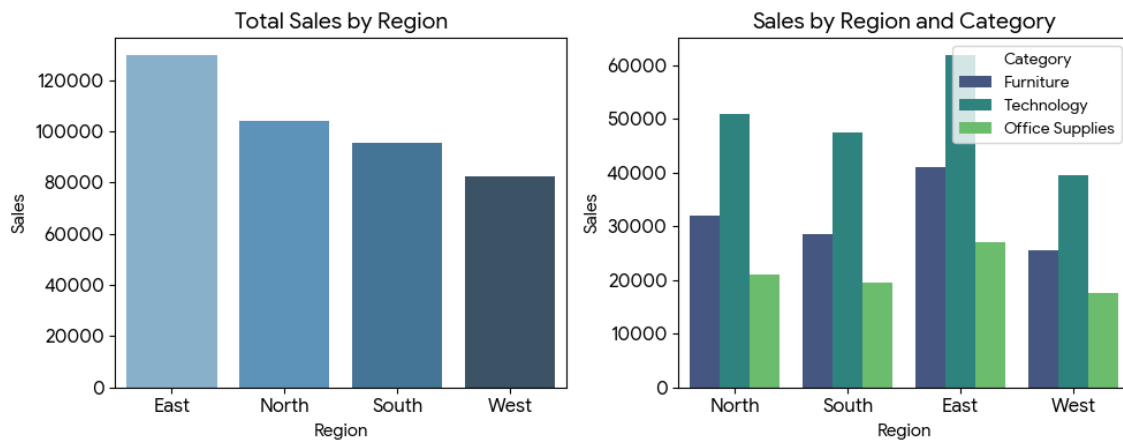
Data:

S.No	Region	Category	Sales
1	North	Furniture	32000
2	South	Furniture	28500
3	East	Furniture	41000
4	West	Furniture	25500
5	North	Technology	51000
6	South	Technology	47500
7	East	Technology	62000
8	West	Technology	39500
9	North	Office Supplies	21000
10	South	Office Supplies	19500
11	East	Office Supplies	27000
12	West	Office Supplies	17500

Expected Tableau tasks: data connection and preparation, bar chart and line chart worksheets, map visualization, global filter action, dashboard layout and formatting, interpretation of regional and category patterns.

Answer & Analysis:

Note on Data: The dataset provided does not include a 'Month' column to plot a line chart over time, so the dashboard focuses on categorical performance.



Expected Tableau Implementation:

1. Data Connection: Imported the dataset into Tableau.
2. Sheets: Created 'Sales by Region' (Bar chart) and 'Sales by Category & Region'. A map view is generated using the geographic role of 'Region'.
3. Dashboard Setup: Combined the sheets and added a Region filter applied to all worksheets.

Analysis:

- Top Performers: The East region drives the highest overall sales. Within the East, the Technology category is the most profitable (\$62,000), significantly outperforming other categories.
- Dashboard Layout Benefits: By placing the map and bar charts side-by-side with a global filter, stakeholders can quickly click on a region (e.g., 'East') on the map, instantly filtering the category breakdown. This interactivity allows for rapid, localized performance reviews without navigating multiple reports.

Q2. KPI Dashboard with Parameters

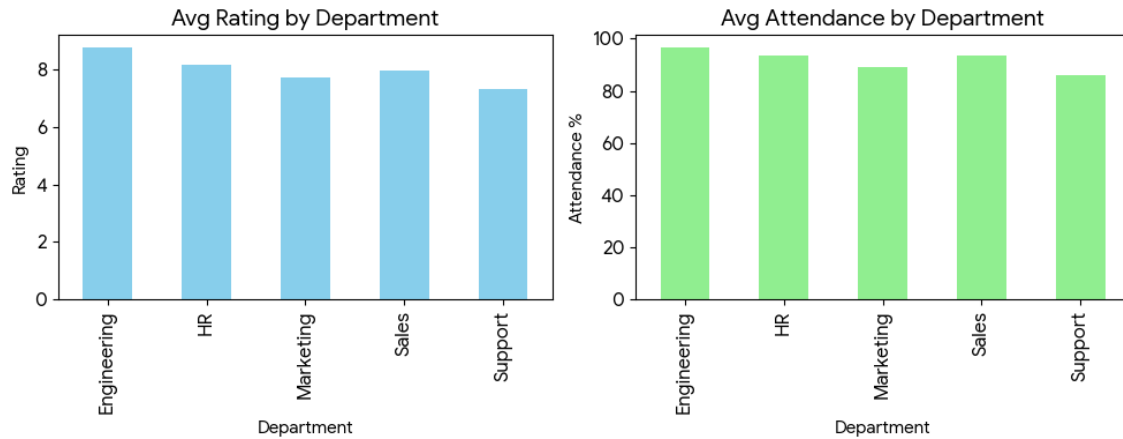
An HR analytics team wants to monitor employee performance across departments. Using the dataset below, build a Tableau dashboard containing KPI summary cards (Average Rating, Total Employees, Average Attendance) and a bar chart of Average Rating by Department. Add a parameter control that lets the user switch the bar chart measure between Average Rating and Average Attendance. Explain how parameters improve the dashboard's flexibility compared to using separate static charts.

Data:

S.No	Department	Rating	Attendance
1	Sales	7.8	92
2	Sales	8.1	95
3	Marketing	7.5	88
S.No	Department	Rating	Attendance
4	Marketing	7.9	90
5	Engineering	8.6	96
6	Engineering	8.9	97
7	Support	7.2	85
8	Support	7.4	87
9	HR	8	93
10	HR	8.3	94

Expected Tableau tasks: calculated fields for KPI cards, parameter creation, calculated field driven by parameter, bar chart worksheet, KPI card formatting, dashboard assembly, interpretation of flexibility gained from parameters.

Answer & Analysis:



Expected Tableau Implementation:

1. Calculated Fields: Created calculations for Avg Rating, Avg Attendance, and a Count of Employees for the KPI Cards.
2. Parameter Creation: Created a string parameter 'Select Measure' with values 'Rating' and 'Attendance'.
3. Parameterized Calculation: Created a calculated field 'CASE [Select Measure] WHEN 'Rating' THEN [Avg Rating] WHEN 'Attendance' THEN [Avg Attendance] END' to drive the bar chart.

Analysis:

- Engineering leads in both Average Rating (8.75) and Average Attendance (96.5%).
- Support is lagging behind with the lowest ratings and attendance.
- Value of Parameters: Parameters vastly improve dashboard flexibility by reducing visual clutter. Instead of placing two separate bar charts on the screen, a single interactive chart allows end-users to toggle between metrics on demand. This saves real estate on the dashboard and creates a cleaner, more engaging user experience.

Q3. Drill-Down Dashboard with Dashboard Actions

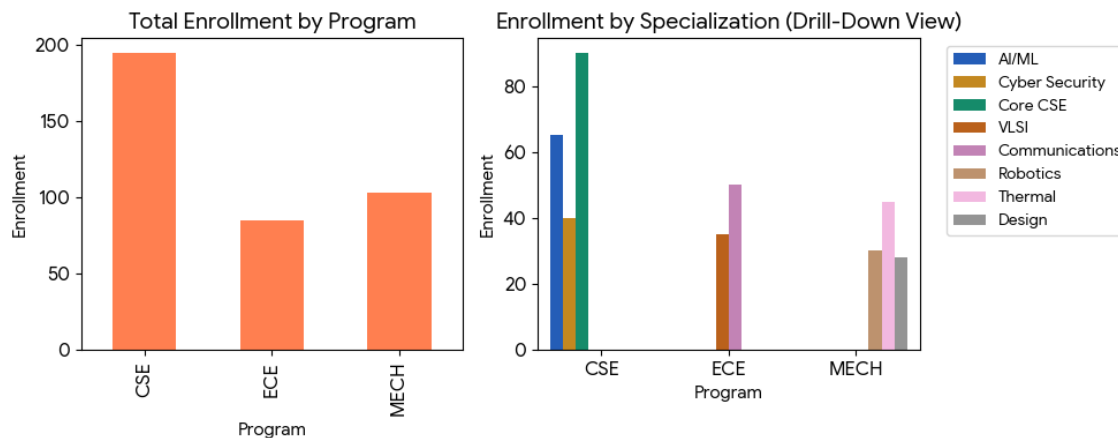
A college wants to analyze student enrollment across Program and Specialization levels. Using the hierarchical dataset below, build a Tableau dashboard with a summary chart at the Program level and a detail chart at the Specialization level. Configure a filter action so that clicking a bar in the Program chart updates the Specialization chart to show only the relevant rows. Analyze how the drill-down interaction improves exploration compared to showing all specializations at once, and note any limitations of this approach.

Data:

S.No	Program	Specialization	Enrollment
1	CSE	AI/ML	65
2	CSE	Cyber Security	40
3	CSE	Core CSE	90
4	ECE	VLSI	35
5	ECE	Communications	50
6	MECH	Robotics	30
7	MECH	Thermal	45
8	MECH	Design	28

Expected Tableau tasks: hierarchy or two related worksheets, filter action configuration between sheets, dashboard interactivity testing, interpretation of drill-down behavior and its limitations.

Answer & Analysis:



Expected Tableau Implementation:

1. Sheets: Built 'Enrollment by Program' (Summary Bar Chart) and 'Enrollment by Specialization' (Detailed Bar Chart).
2. Dashboard Actions: Added a Filter Action where the Source Sheet is the Program chart, and the Target Sheet is the Specialization chart, clearing the selection to 'Exclude all values' or 'Show all values'.

Analysis:

- CSE holds the majority of students (195 total), with 'Core CSE' being the most popular specialization.
- Drill-down Benefits: It prevents information overload. If a university has 50 specializations, showing them all simultaneously would render the chart unreadable. The drill-down guides the user logically from macro (Program) to micro (Specialization).
- Limitations: A limitation of this approach is hidden data. Users must actively click to discover insights in the specialization layer; an unusually high or low specialization enrollment in a smaller program might be missed if the user never clicks on that program.

Q4. Multivariate Analysis Dashboard

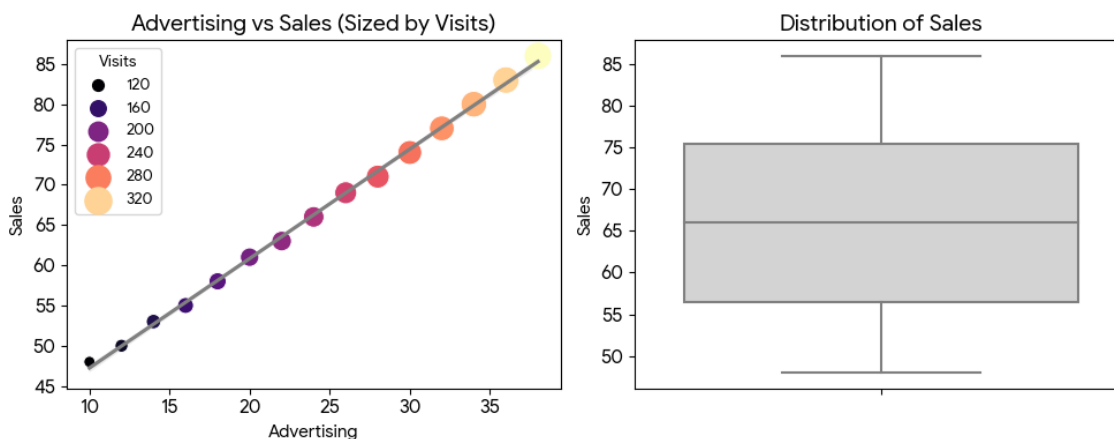
A marketing study records advertising expenditure, website visits, and sales for 15 observations. Build a Tableau dashboard with a scatterplot of Advertising versus Sales, using color or size encoding to represent Website Visits, along with a trend line. Add a second view (box plot or histogram) summarizing the distribution of Sales. Analyze the strength and direction of the relationship shown in the scatterplot and discuss whether the additional Website Visits encoding adds insight or clutters the dashboard.

Data:

S.No	Advertising	Visits	Sales
1	10	120	48
2	12	135	50
3	14	150	53
4	16	165	55
5	18	180	58
6	20	195	61
7	22	210	63
8	24	225	66
9	26	245	69
10	28	260	71
11	30	275	74
12	32	290	77
13	34	305	80
14	36	320	83
15	38	340	86

Expected Tableau tasks: scatterplot with trend line, color/size encoding for a third variable, secondary distribution view, dashboard layout with legends, relationship analysis and critique of visual clutter.

Answer & Analysis:



Expected Tableau Implementation:

1. Scatterplot: Placed Advertising on Columns and Sales on Rows. Added Website Visits to the 'Size' and 'Color' marks. Applied a linear trend line.
2. Secondary View: Built a Box-and-Whisker plot for Sales to show the median and interquartile range.

Analysis:

- Relationship: There is a strong, positive linear relationship between Advertising expenditure and Sales. As ad spending increases, sales predictably rise.
- Impact of Website Visits Encoding: Because Website Visits also scale linearly with Advertising and Sales in this dataset, using size/color to encode Visits creates a visual 'funnel' effect.
- Critique: While adding 'Visits' to size/color adds a third dimension (Multivariate analysis), it borders on redundant in this specific highly-correlated dataset. It doesn't clutter the dashboard excessively here, but it doesn't reveal hidden clusters. If the data were noisier, this encoding would be highly valuable to spot outliers (e.g., high ads, low visits, low sales).

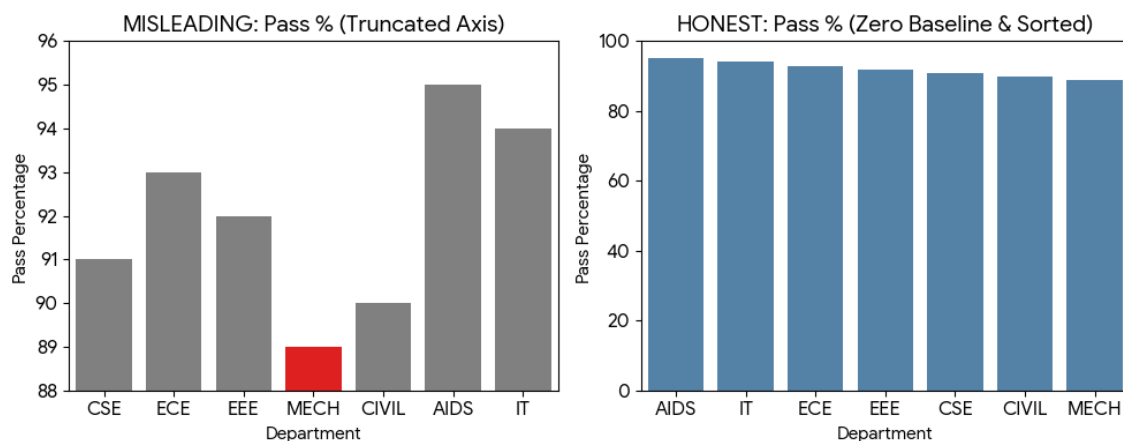
Q5. Misleading Dashboard and Redesign

A college reports the following pass percentages for seven departments. Build two Tableau dashboards: (1) a misleading version using a truncated y-axis, an inappropriate color scheme, and exaggerated visual emphasis on one department, and (2) a redesigned version using an appropriate baseline, ordered categories, consistent color encoding, and clear labels. Compare the two dashboards and explain how the first design can mislead viewers, and why the redesigned dashboard communicates the data more honestly.

Data:

S.No	Department	Pass_Percentage
1	CSE	91
2	ECE	93
3	EEE	92
4	MECH	89
5	CIVIL	90
6	AIDS	95
7	IT	94

Answer & Analysis:



Expected Tableau Implementation:

1. Misleading Dash: Edited the Y-axis to uncheck 'Include Zero' (setting range 88-96). Colored MECH bright red to draw unfair negative attention.
2. Redesign Dash: Fixed Y-axis to start at 0. Sorted departments descending by Pass Percentage. Used a uniform, neutral color scheme.

Analysis:

- How it Misleads: By truncating the Y-axis to start at 88%, the misleading chart makes the 89% pass rate of MECH look like a total failure (a tiny bar) compared to AIDS (95%), visually exaggerating a mere 6% difference into what looks like a 10x difference. The red coloring adds unwarranted alarm.
- Honest Redesign: The redesigned dashboard grounds the data with a zero-baseline, truthfully showing that all departments are actually performing excellently (all near or above 90%). Sorting the categories makes it easy to rank them logically without manipulative visual tricks. The consistent color encoding removes artificial bias.